BASIC
ELECTRICITY AND ELECTRONIC COMPONENTS

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CONTENT

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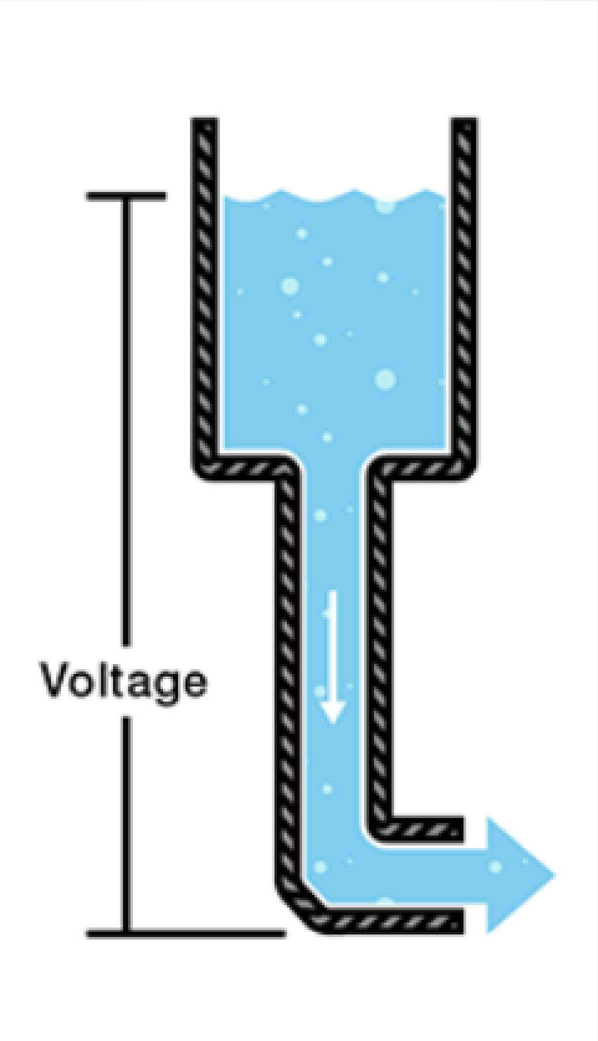
3. Kirchhoff’s law
4. Electronic switch (MOSFET)
5. Lab1: check electrical characteristic.
6. Lab2: om’s law, Kirchhoff’s law.
7. Lab3: using MOSFET as switch

VOLTAGE

Voltage: (The pressure)
Voltage, measured in volts (V), is the electrical "pressure" that pushes charged electrons through a conductive loop. It is the difference in electric potential energy between two points.



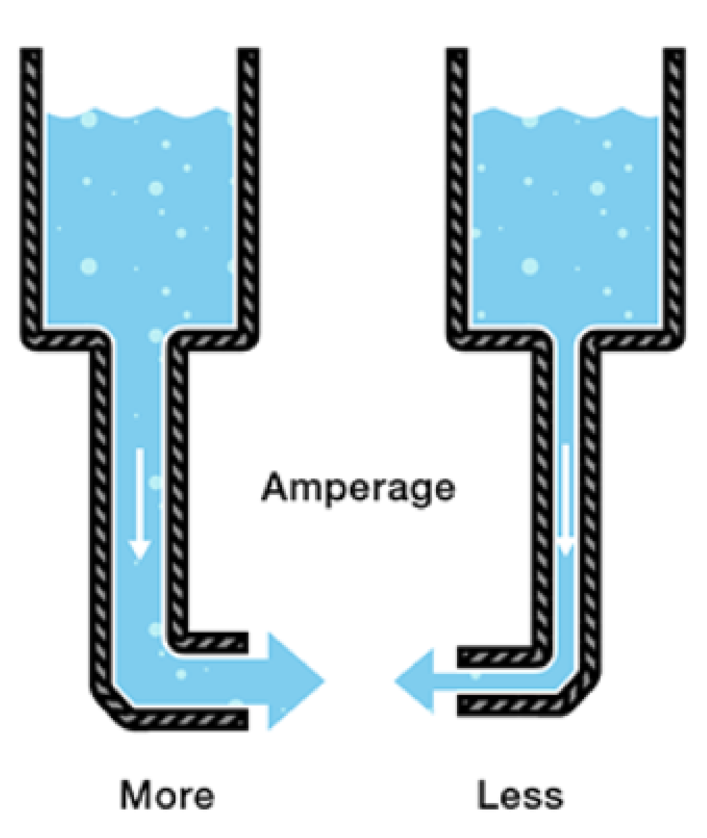
Analogy: The water pressure in a hose. If you have a water tank on a high hill (high voltage), the water pressure will be higher than if the tank is on a low hill (low voltage)

CURRENT

Current: The flow

Current, measured in amperes or amps (A), is the rate of flow of electric charge (electrons) through a circuit. The higher the current, the more electrons are flowing per second.

Analogy: The volume of water flowing through the hose. If you open the hose's valve wider, the rate of water flow increases, just as a lower resistance in an electrical circuit allows for a higher flow of current.



POWER

Power: The work rate

Power, measured in watts (W), is the rate at which electrical energy is used or produced. It is the product of voltage and current ($P = V \times I$). A device with high power does more work per unit of time.

Analogy: The rate at which the flowing water does work. A high-pressure, high-flow fire hose can knock things over much faster than a garden hose, meaning it has a higher power output.

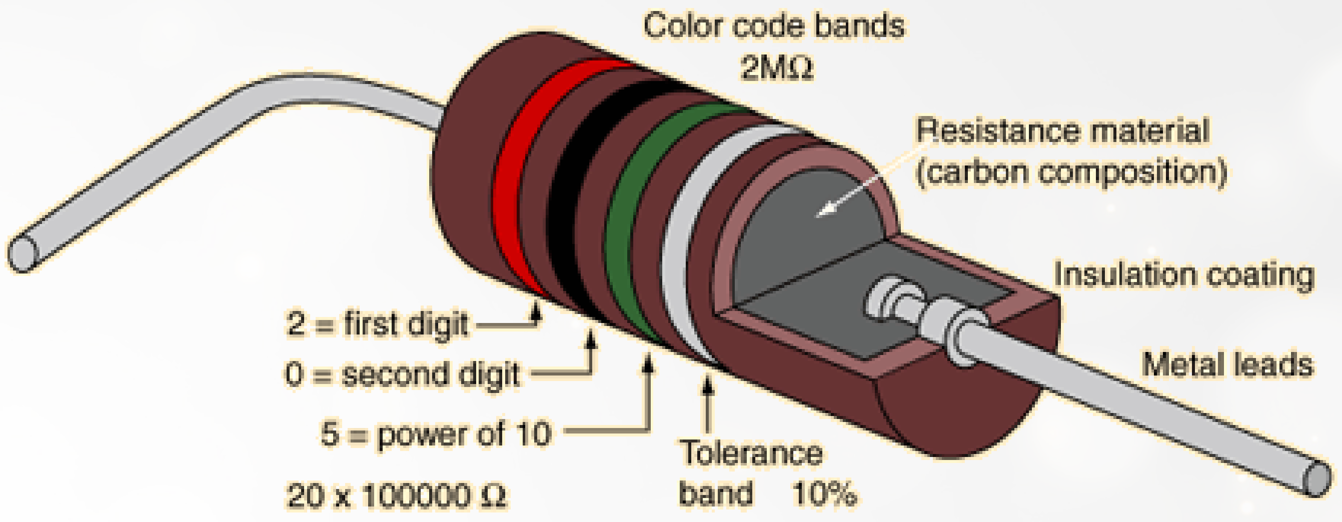
ENERGY

Energy: The total work

Energy, measured in joules (J) or kilowatt-hours (kWh), is the total amount of work done over a period of time. While power is a rate, energy is the cumulative result.

Analogy: The total amount of water that has flowed out of the hose. Using the high-power fire hose for one minute would use far more total water (energy) than using a low-power garden hose for one minute, even though the time is the same.

RESISTOR



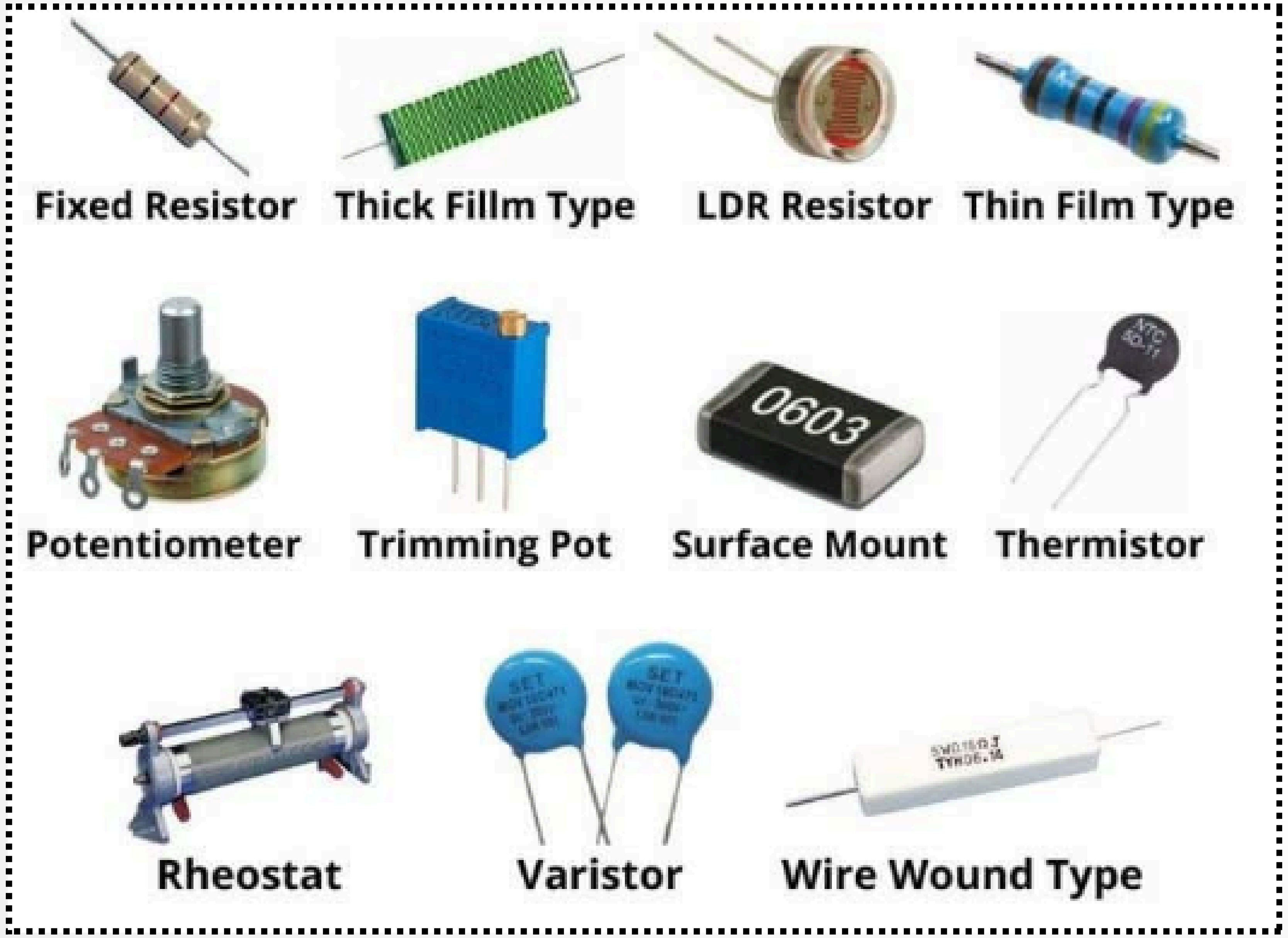
Resistor: Limiting the flow

A resistor is a passive component that resists the flow of electric current. It converts electrical energy into heat and is used to limit and regulate the amount of current in a circuit.

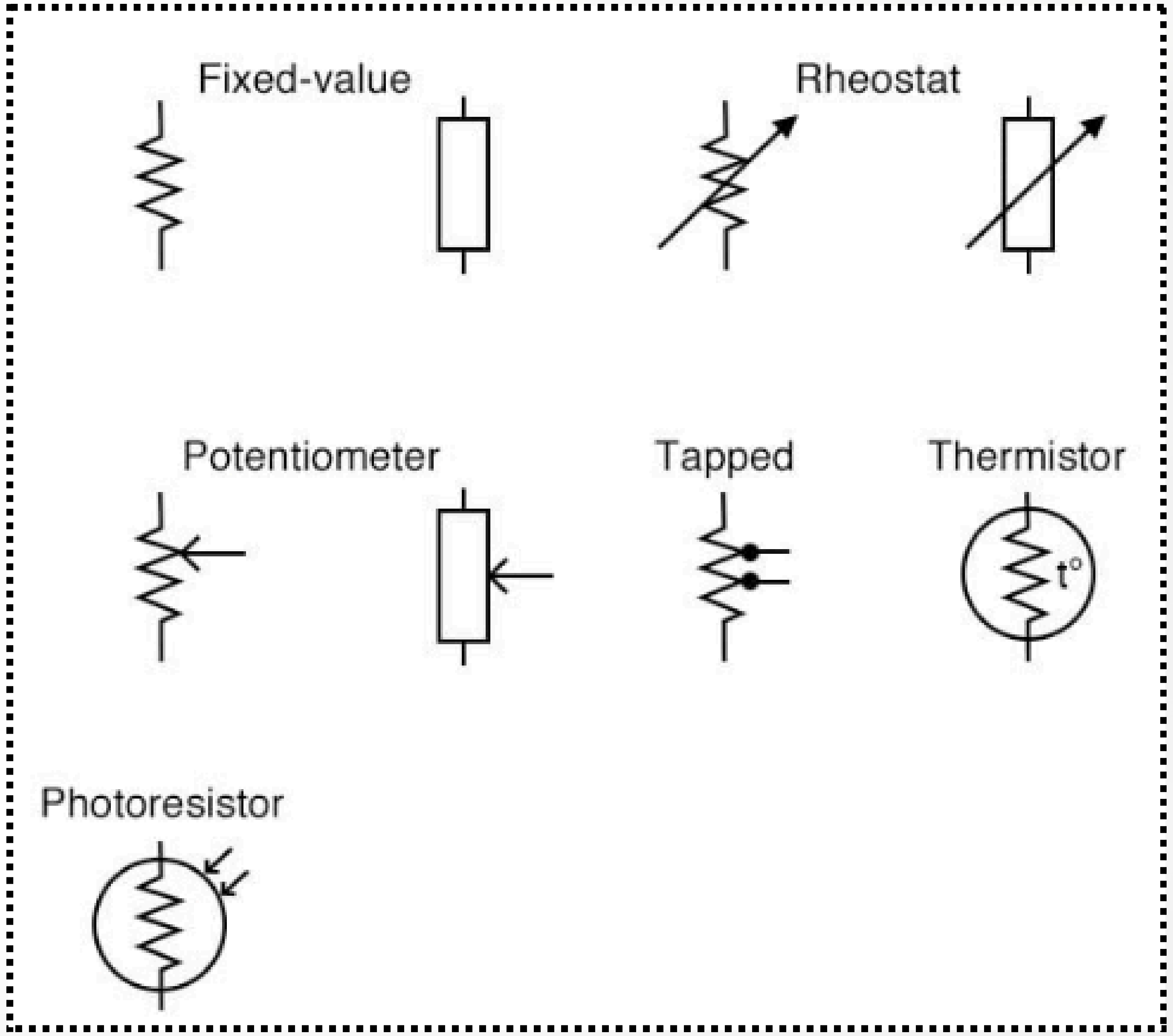
- **Water analogy:** A resistor is like a constriction in a water pipe. The narrower the pipe, the more it resists the flow of water.
- **Microscopic view:** A resistor's material is designed to impede the movement of electrons. The electrons collide with atoms as they pass through, causing friction that produces heat and limits the current.

RESISTOR

Resistor in application

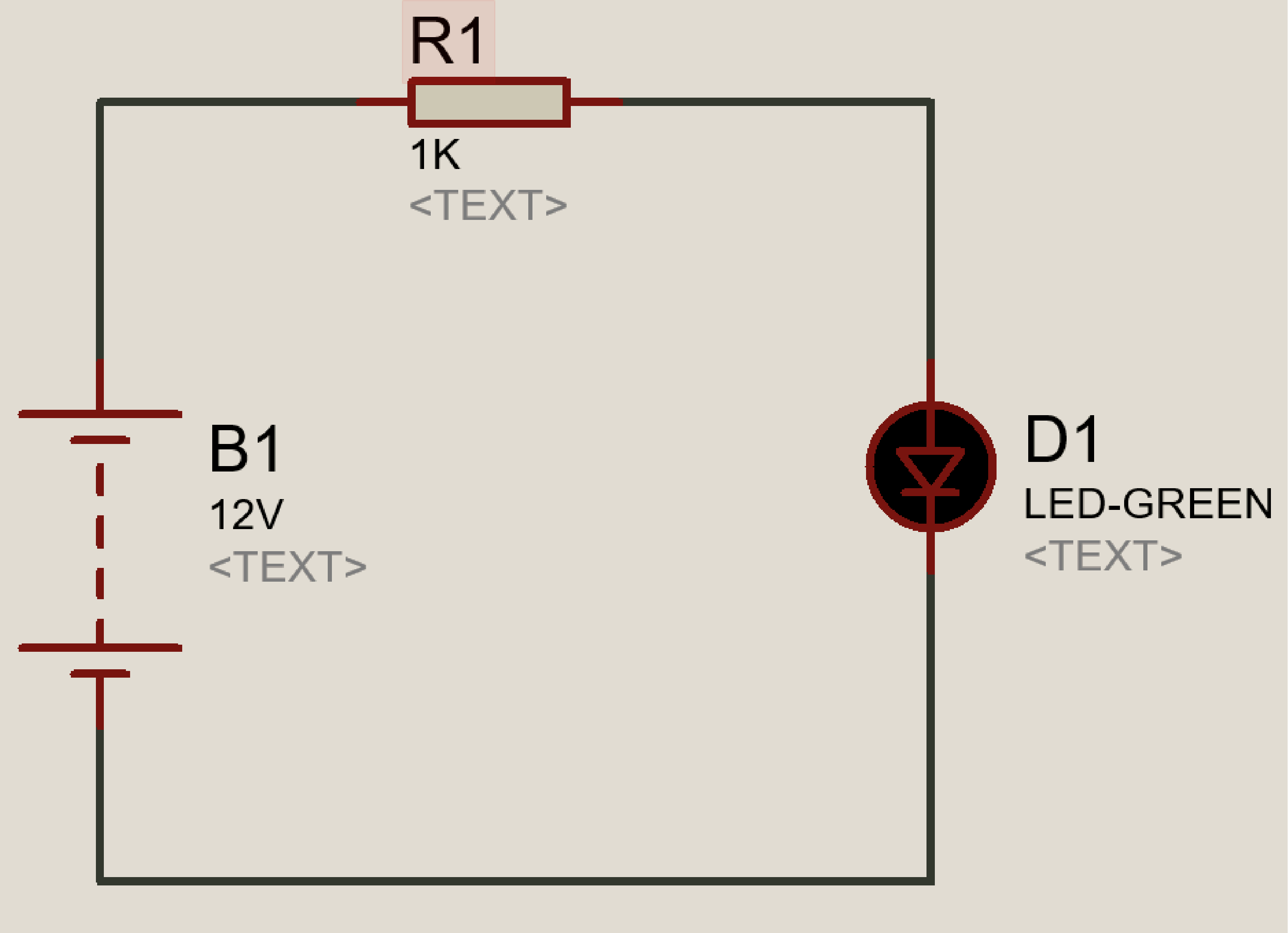


Resistor circuit symbol



RESISTOR

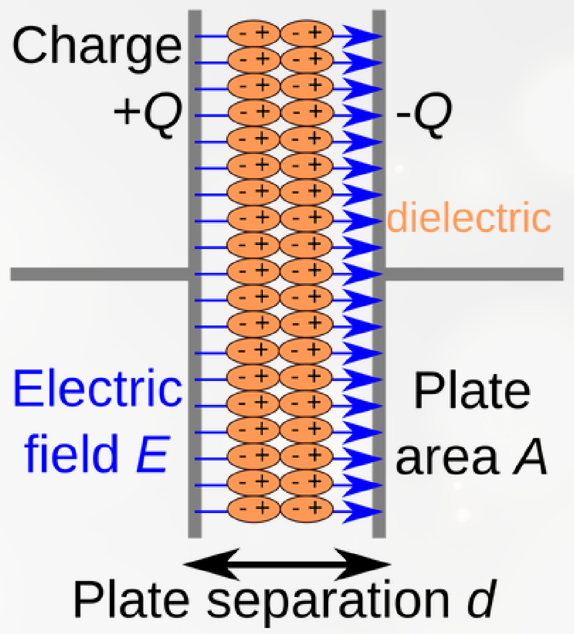
Example: calculate the current flow in circuit. check if we change the value of resistor R1.



Solution : for solving this circuit we need to use Om’s Law:

$$V = R.I$$

CAPACITOR



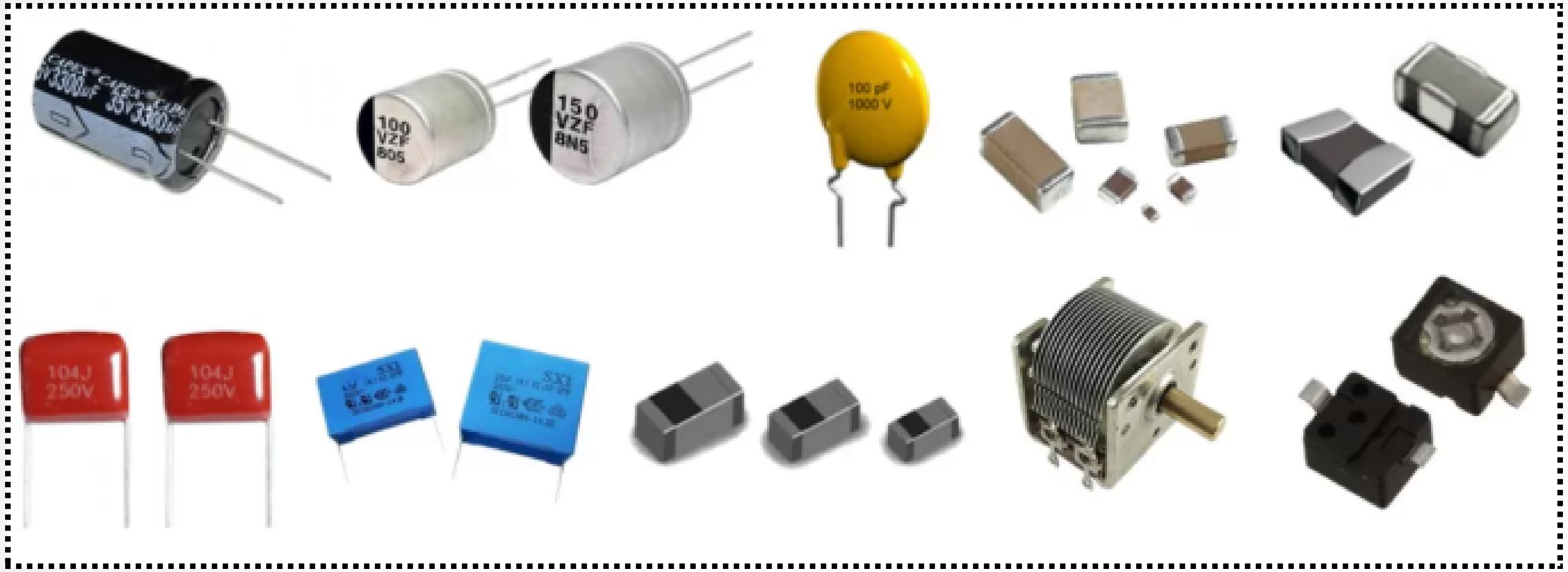
Capacitor: Storing charge

A capacitor is a passive component that stores electrical energy in an electric field. It consists of two conductive plates separated by an insulating material called a dielectric. It resists sudden changes in voltage.

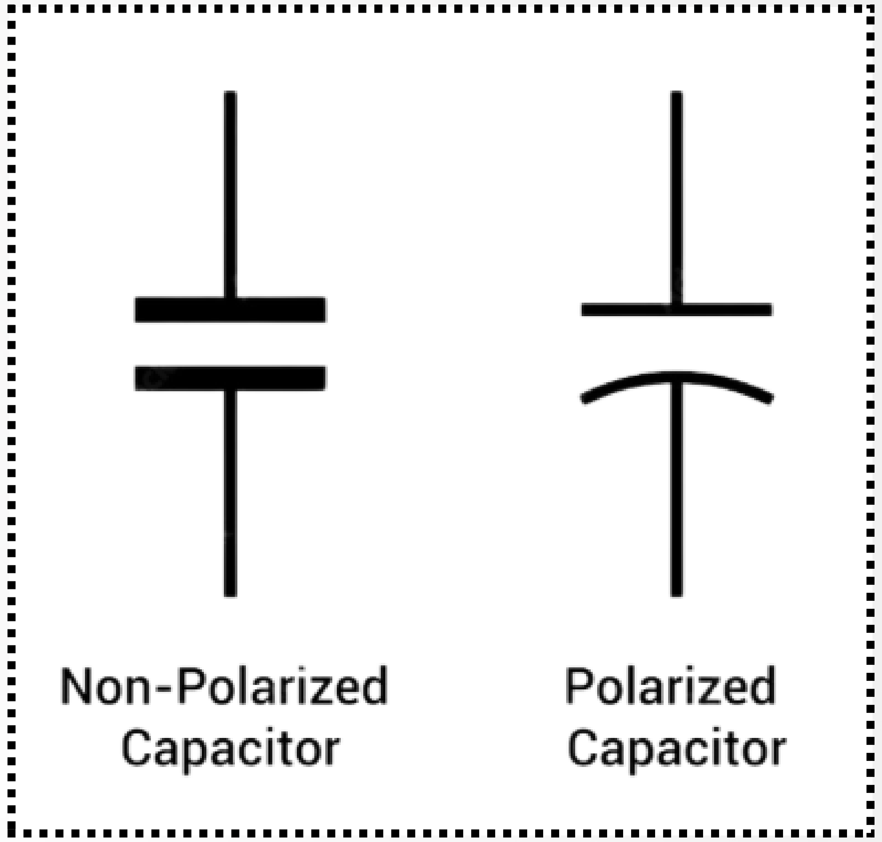
Water analogy: A capacitor is like an elastic diaphragm inside a pipe. It stores energy by stretching as water pressure increases and releases it as pressure decreases, allowing water to flow even when the pressure source is removed.

CAPACITOR

Capacitor in real



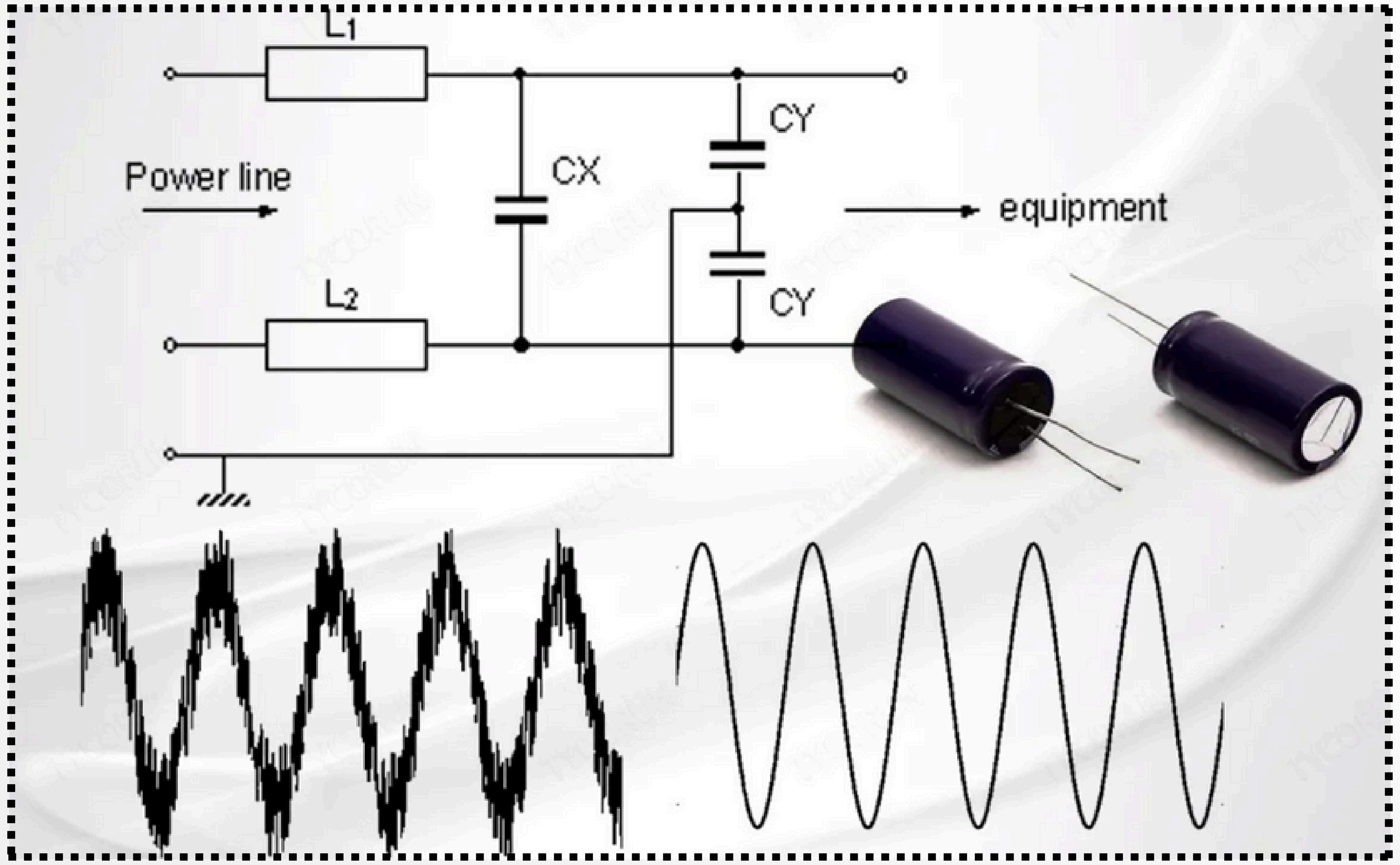
Capacitor symbol

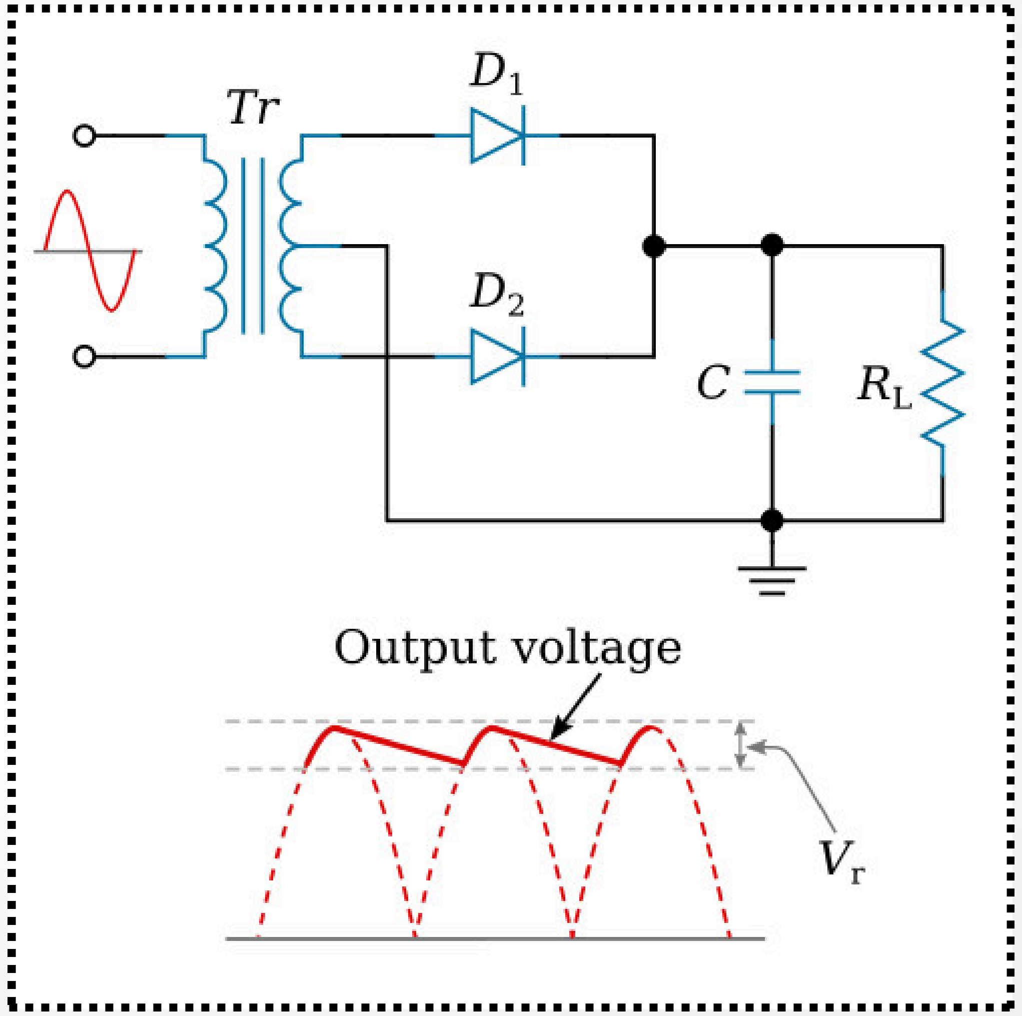


CAPACITOR

Capacitor usage

- **Filtering**: A capacitor can smooth out voltage fluctuations in a power supply, converting pulsating DC into a smoother, more stable output.



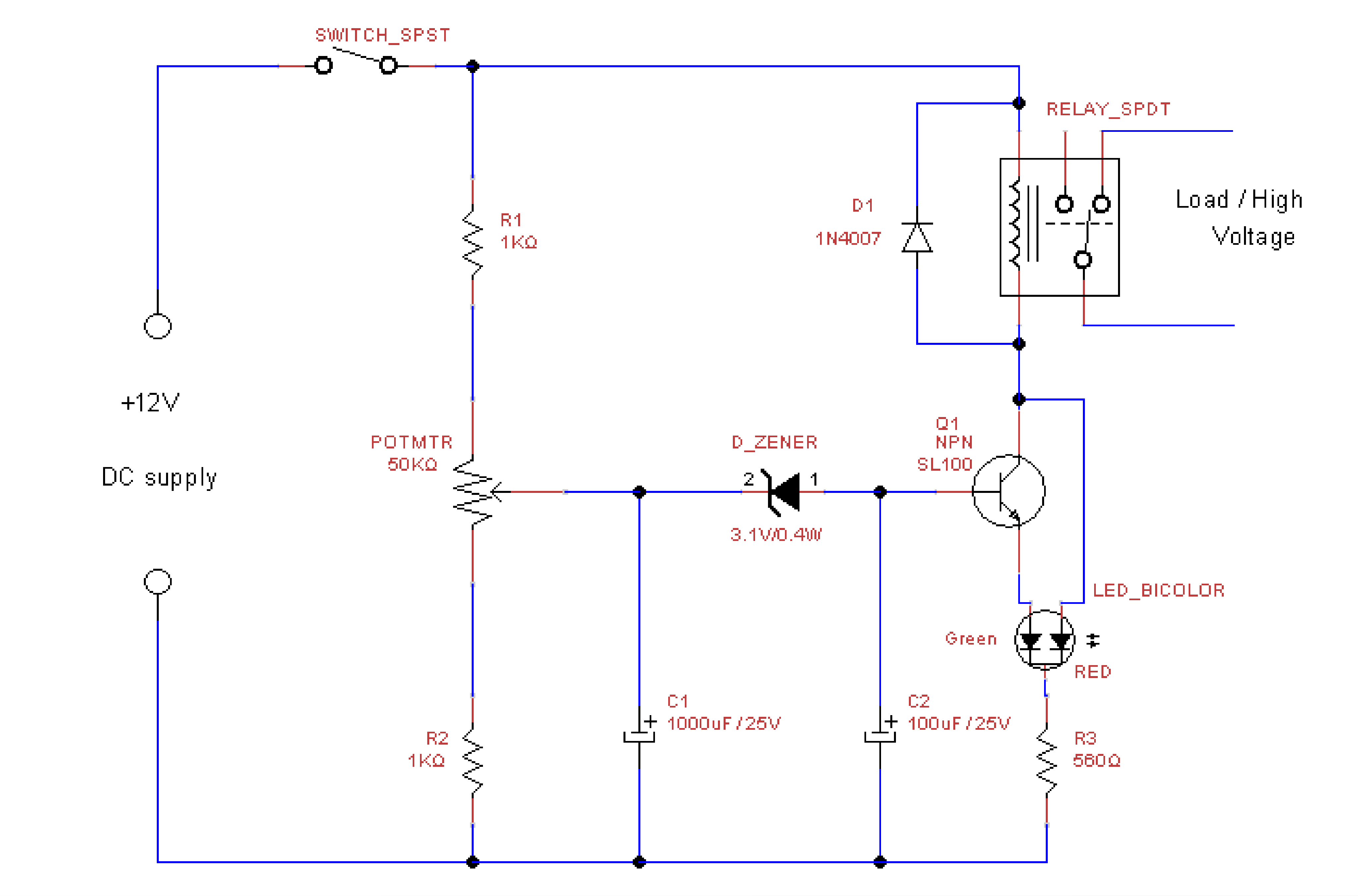


CAPACITOR

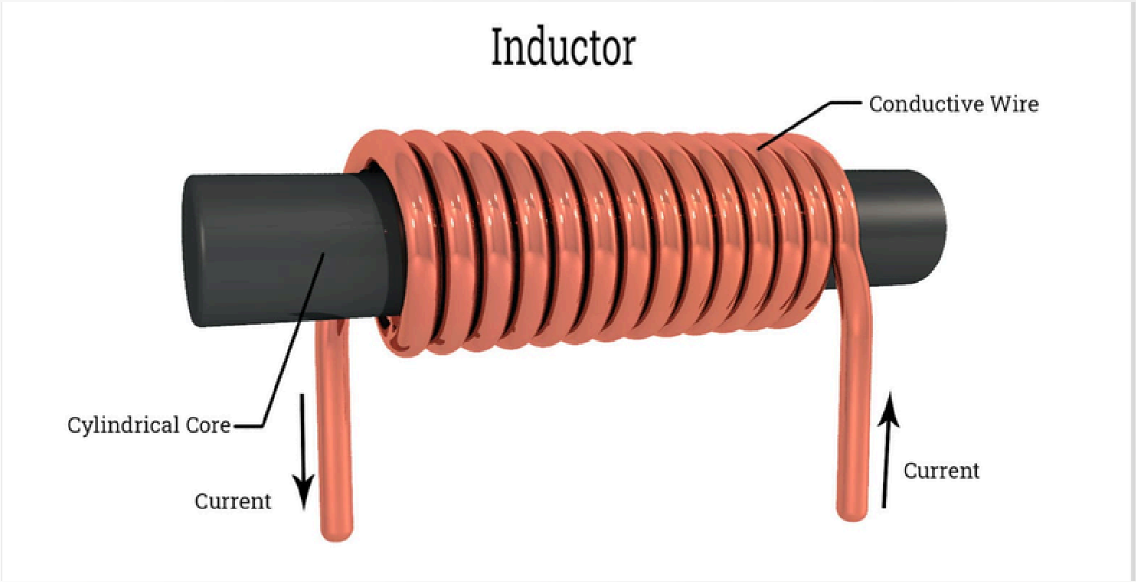
Capacitor usage

- **Timing:** A resistor and capacitor can be used together to create a timer, as the time it takes for the capacitor to charge and discharge is predictable.

$$V_C(t) = V_0 \cdot (1 - e^{-t/\tau})$$



INDUCTOR



Inductor: Opposing change

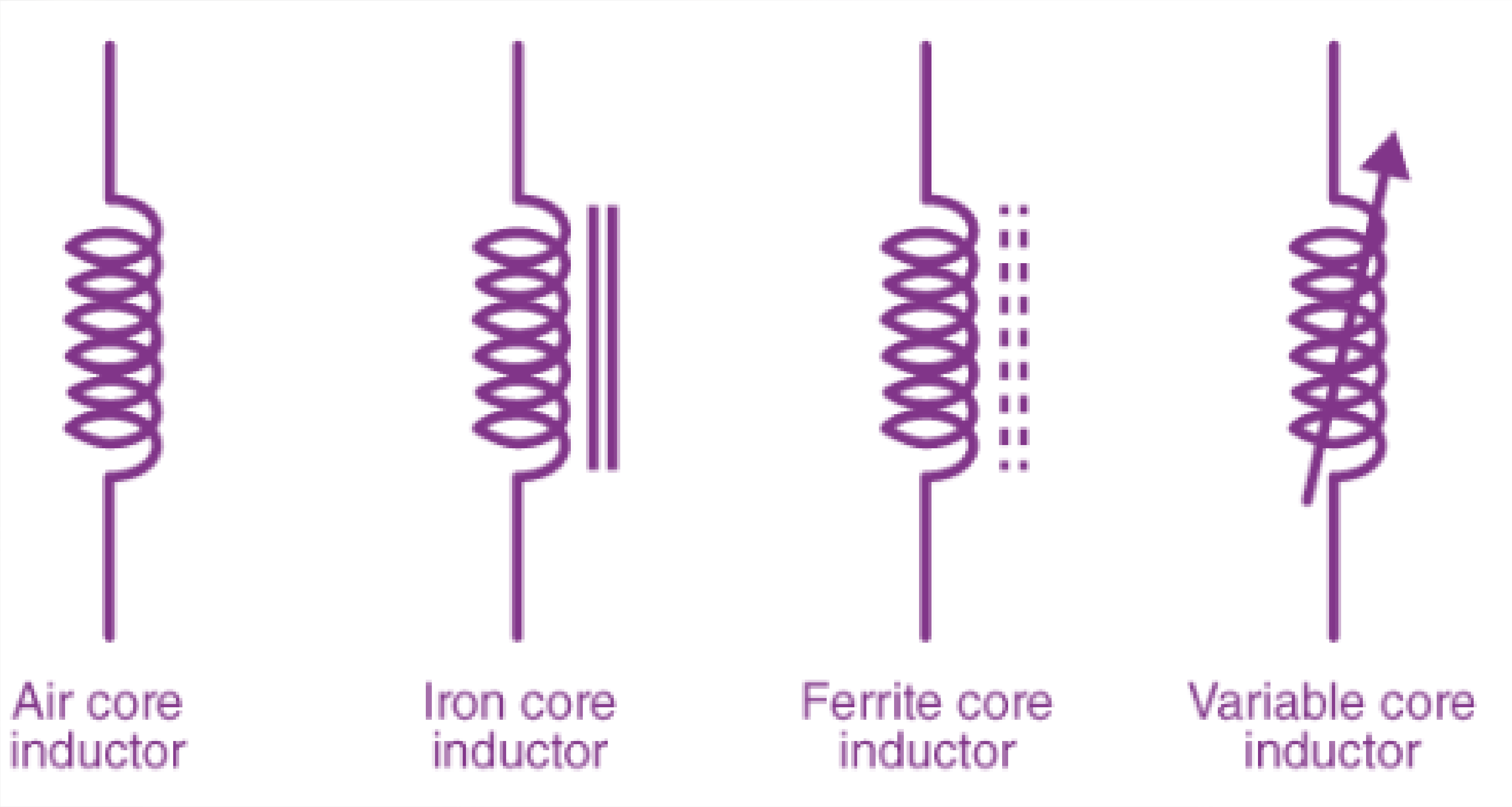
An inductor is a passive component that stores energy in a magnetic field when an electric current flows through it. Typically, it's a coil of wire that opposes changes in current.

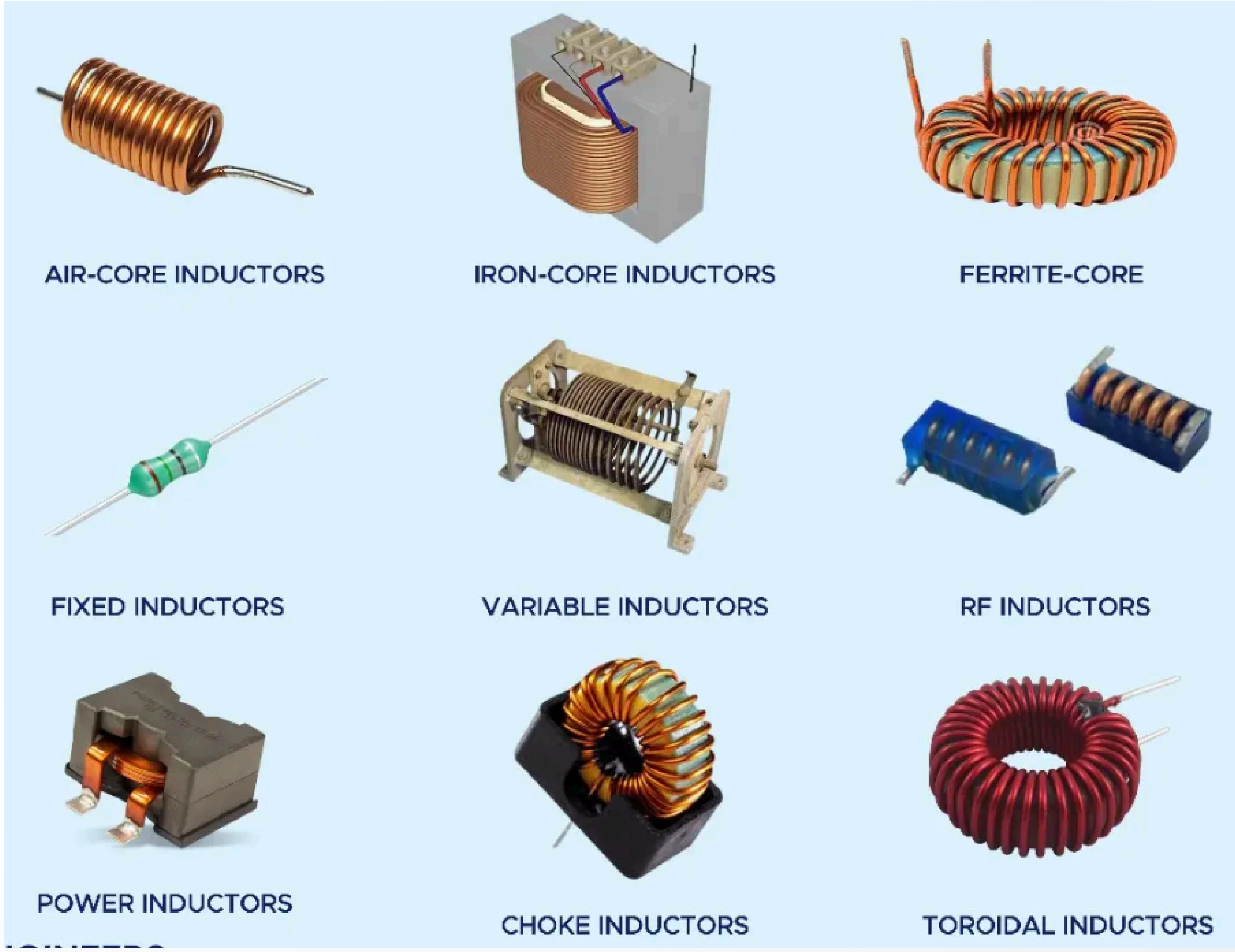
Water analogy: An inductor can be visualized as a water wheel. It resists sudden changes in water flow; it takes time and energy to get it spinning, but once it's in motion, it resists stopping and will continue to push water forward for a time.

INDUCTOR

inductor type in application

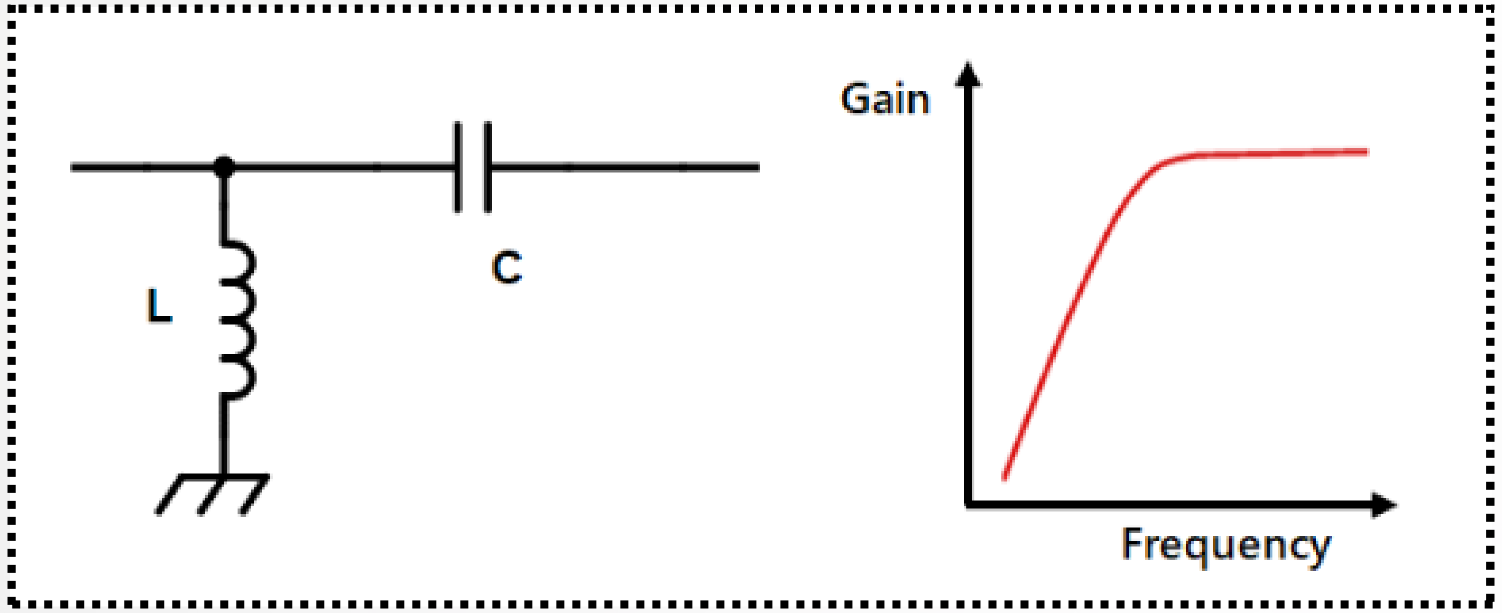
inductor symbol

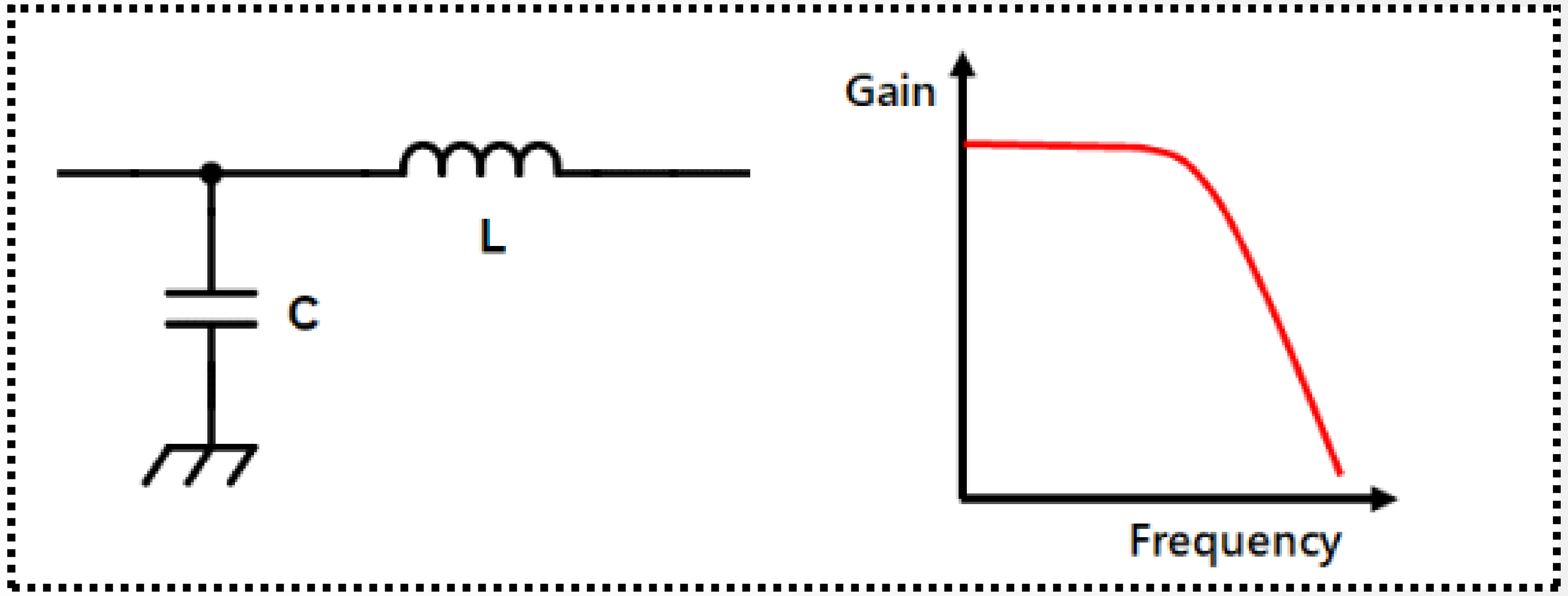




INDUCTOR

- **Filters**: An inductor is used with a capacitor to create filters that block certain frequencies while allowing others to pass.

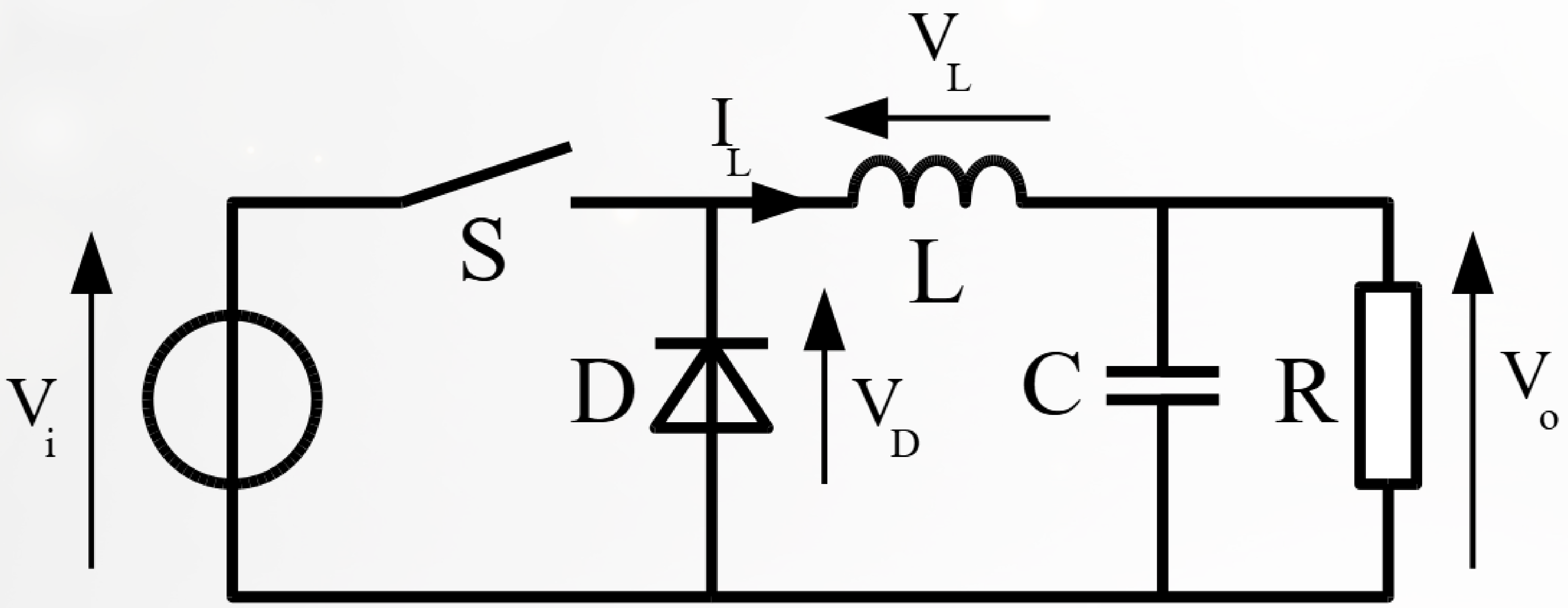




- Low pass Filter
- high pass Filter

INDUCTOR

Power supplies: In switched-mode power supplies, inductors store and transfer energy to produce a DC output.



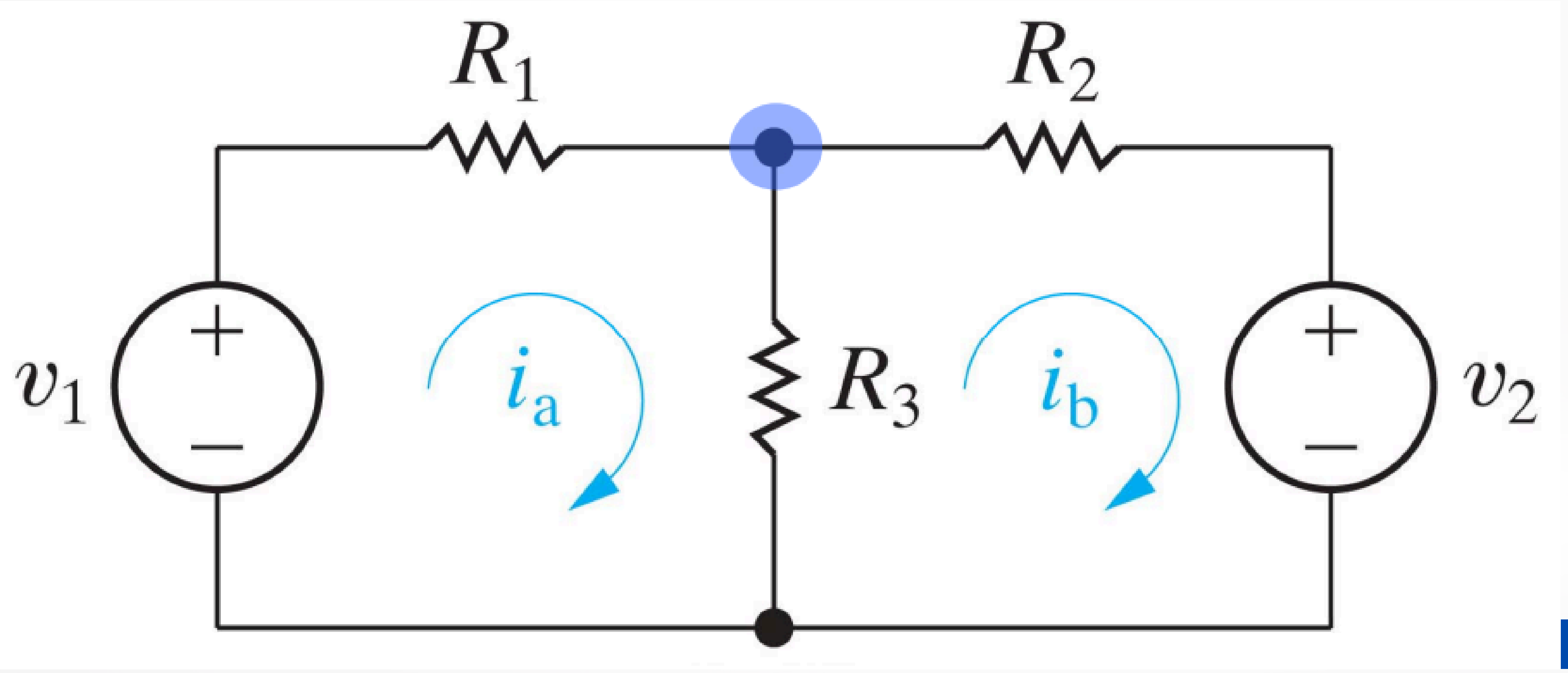
DC-DC converter

$$Vo = Vi \times D$$

D: Duty ratio

KIRCHHOFF’S LAW

Kirchhoff's laws are two fundamental principles used in electrical engineering to analyze complex circuits that cannot be simplified using series and parallel resistance calculations. They are derived from the conservation of charge and energy.



KIRCHHOFF'S LAW

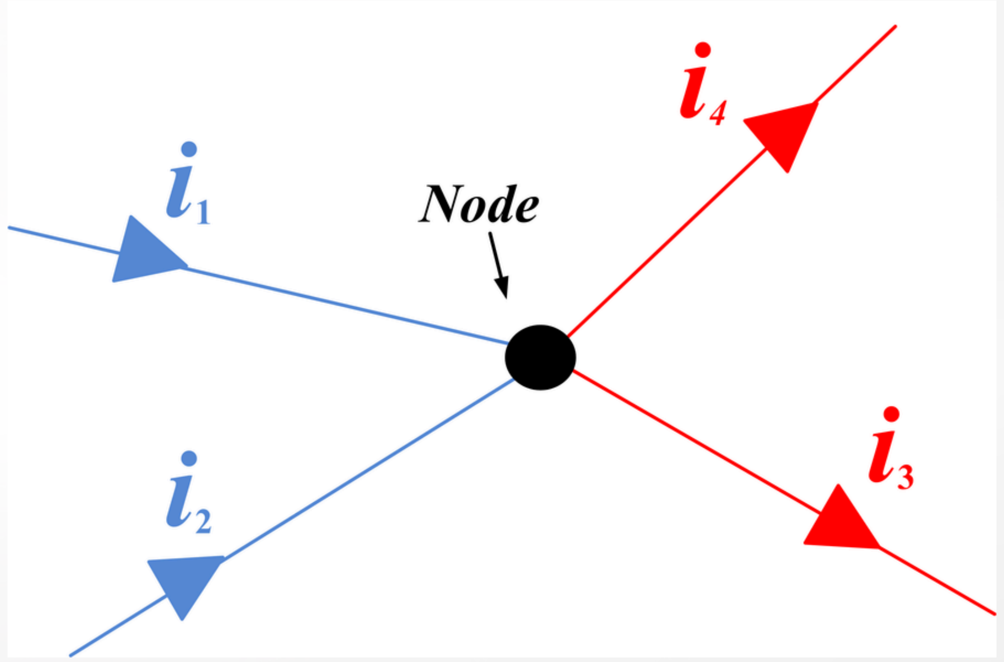
1. Kirchhoff's Current Law (KCL)

Also known as the junction rule, KCL is based on the principle of conservation of electric charge.

What it states:

The total current entering a junction (or node) in an electrical circuit is equal to the total current leaving that junction.

$$\Sigma i_{in} = \Sigma i_{out}$$



KIRCHHOFF'S LAW

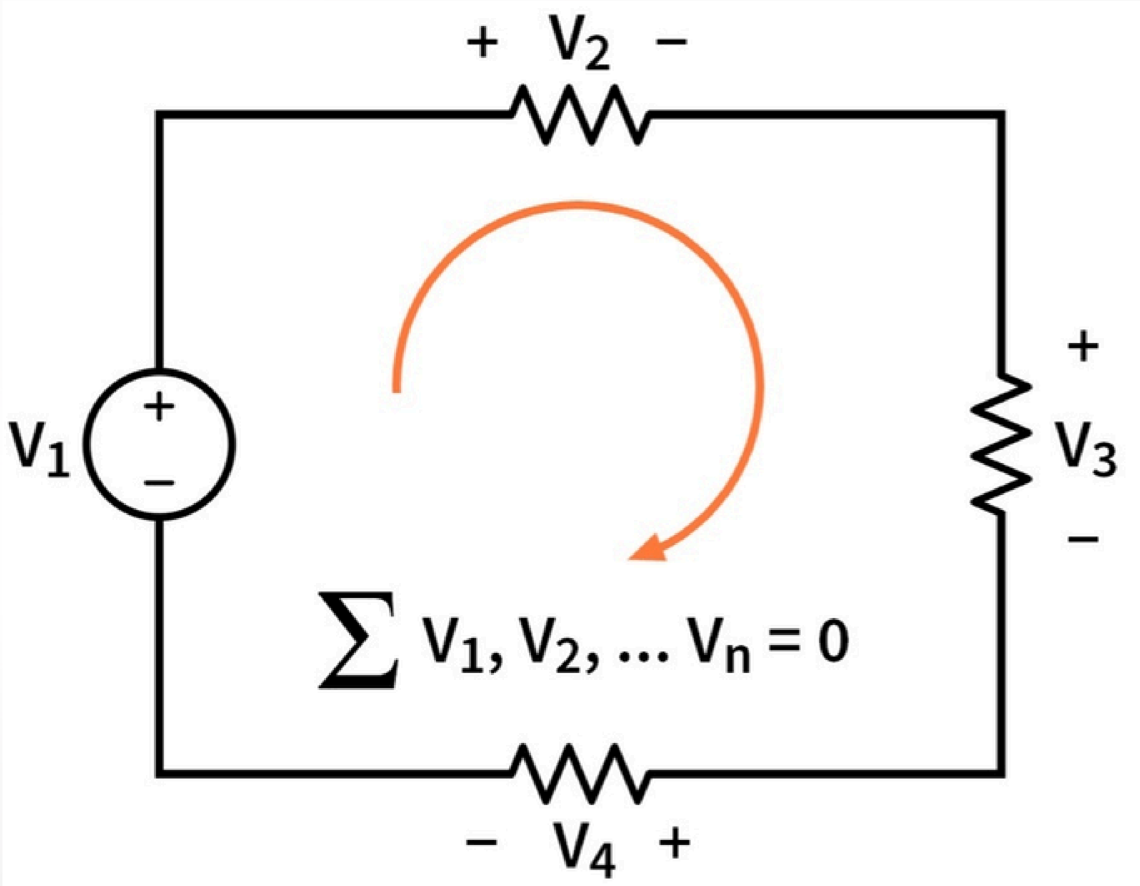
2. Kirchhoff's Voltage Law (KVL)

Also known as the loop rule, KVL is based on the principle of conservation of energy.

What it states:

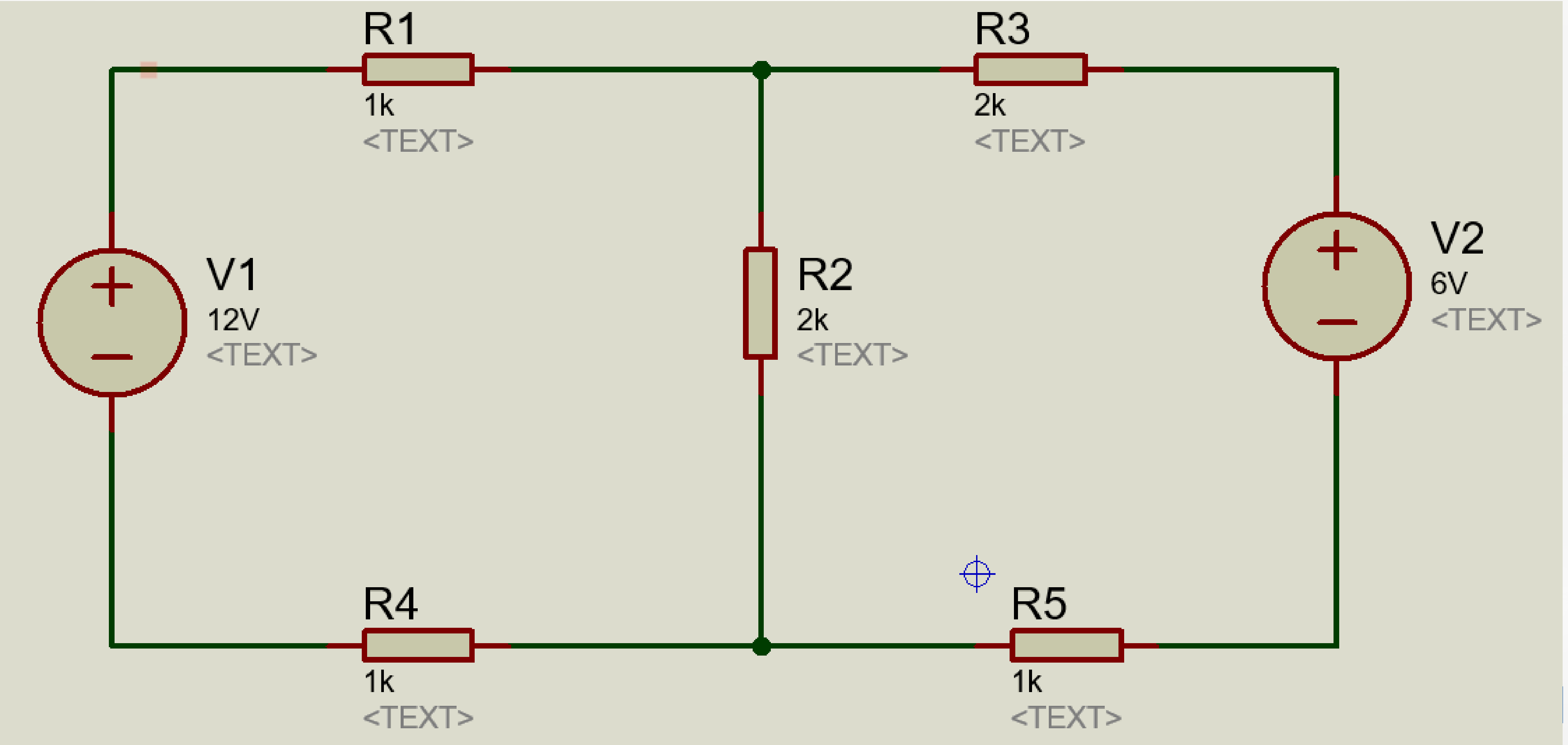
The algebraic sum of all the potential differences (voltages) around any closed loop in a circuit is zero. This means that any energy supplied by a voltage source must be equal to the energy dissipated or stored by the components in the loop.

$$\Sigma V_{loop} = 0$$



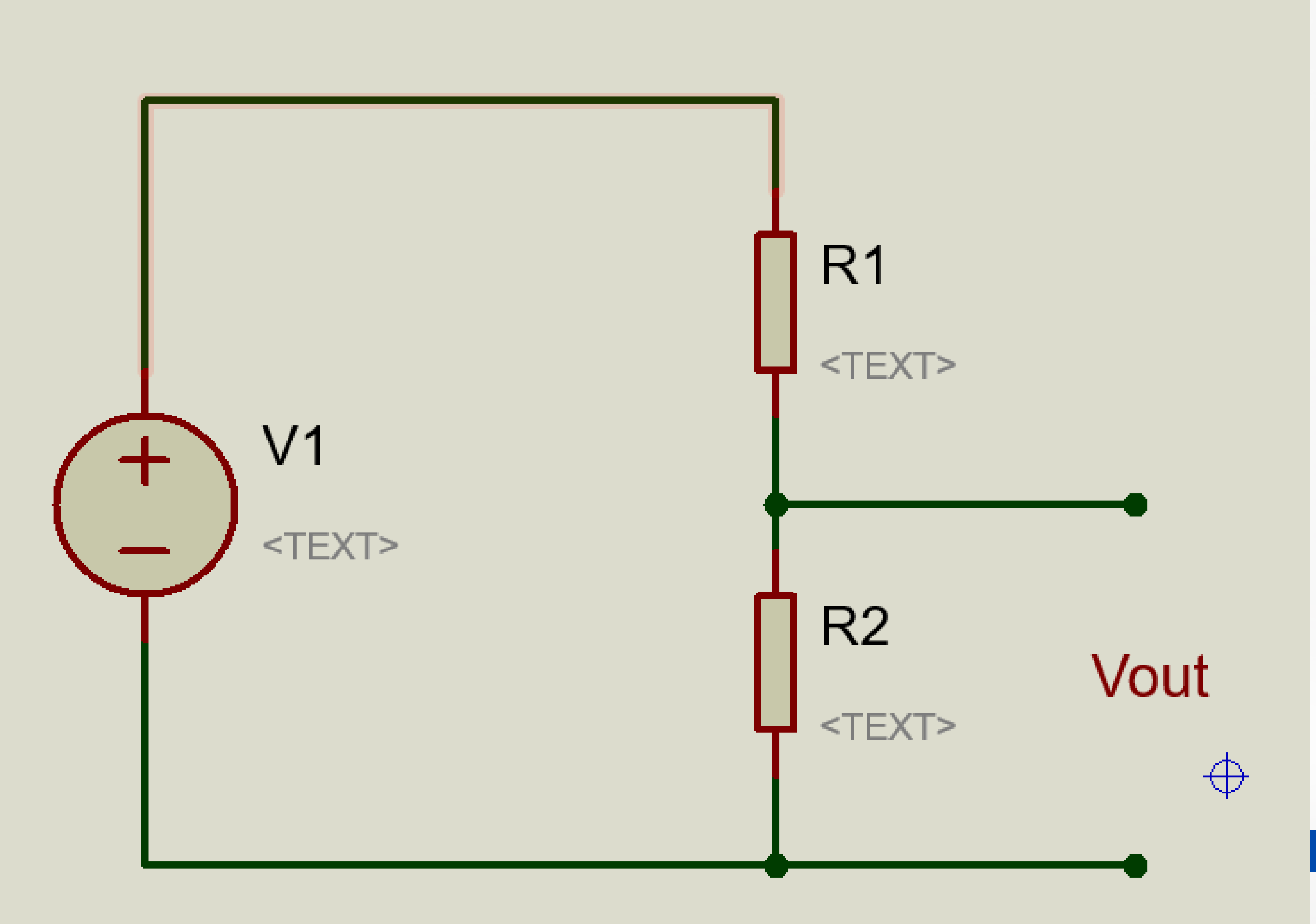
KIRCHHOFF'S LAW

Example : let's find the current cross resistor R3 and voltage drop on R2.



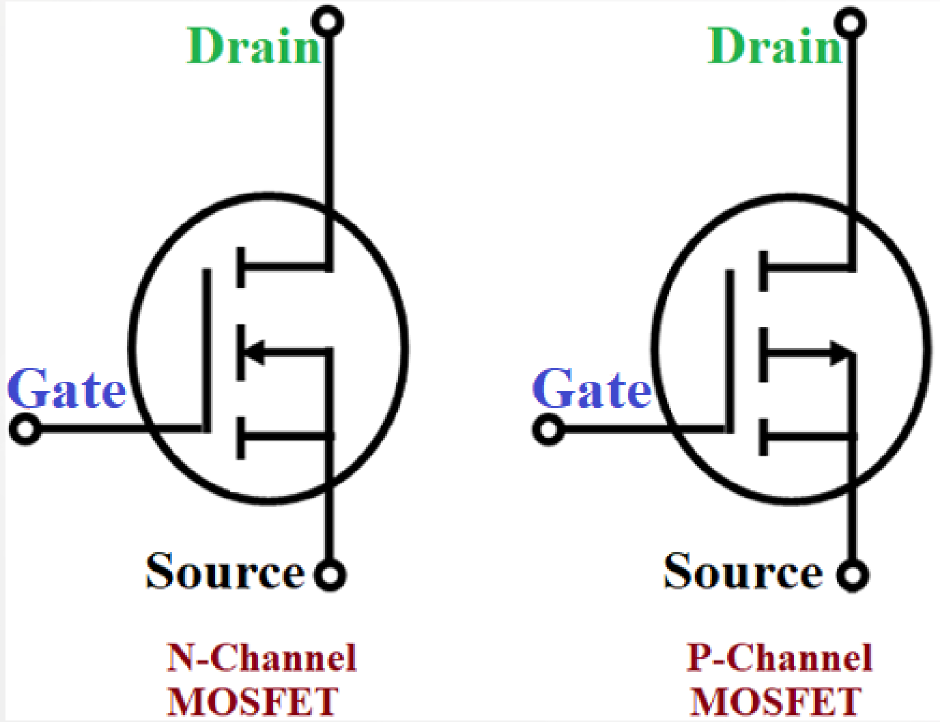
KIRCHHOFF'S LAW

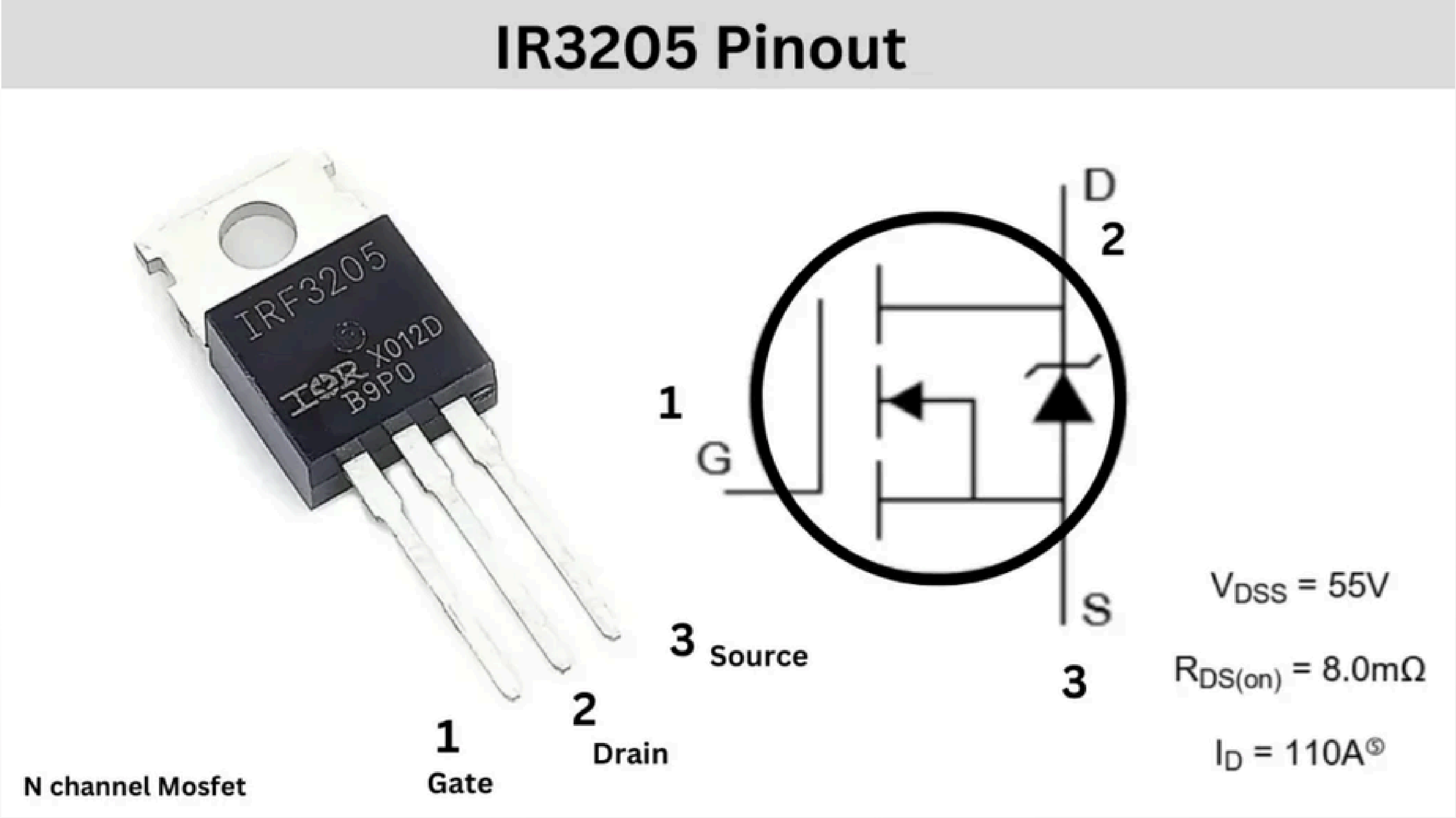
Example : show that $$Vout = V1\left(\frac{R2}{R1+R2}\right)$$



ELECTRONIC SWITCH

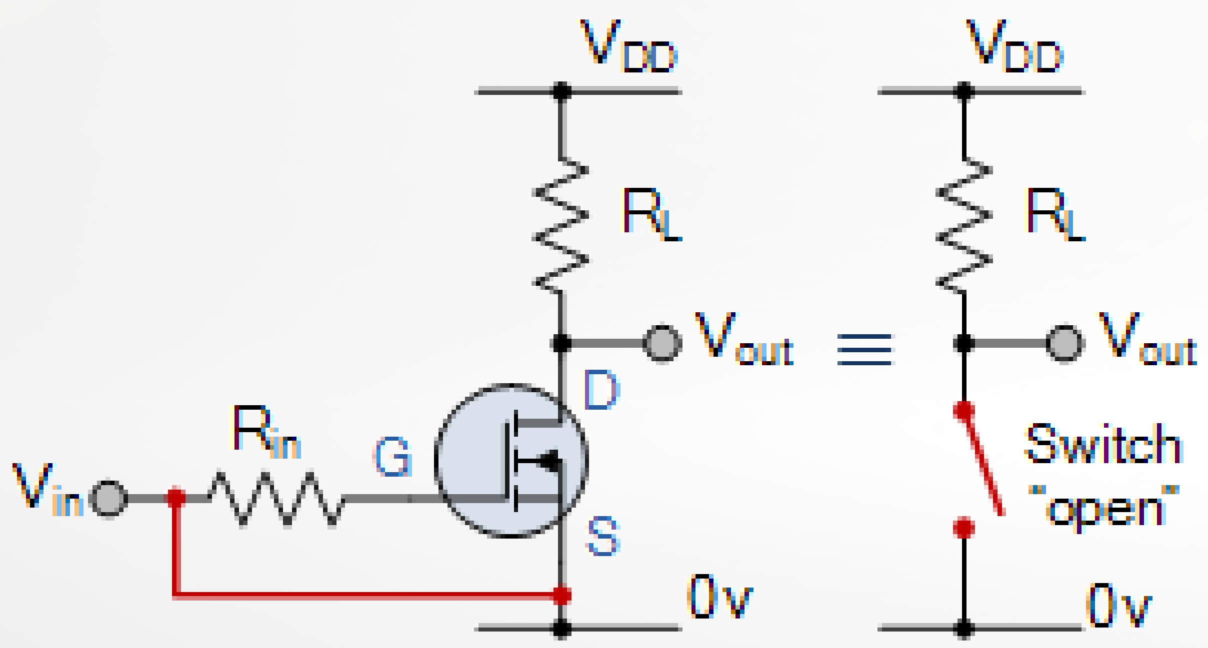
A MOSFET, or **Metal-Oxide-Semiconductor Field-Effect Transistor**, is a type of electronic switch that uses a voltage to control the flow of current. It is the most common transistor in both digital and analog circuits, found in devices from smartphones to microprocessors.

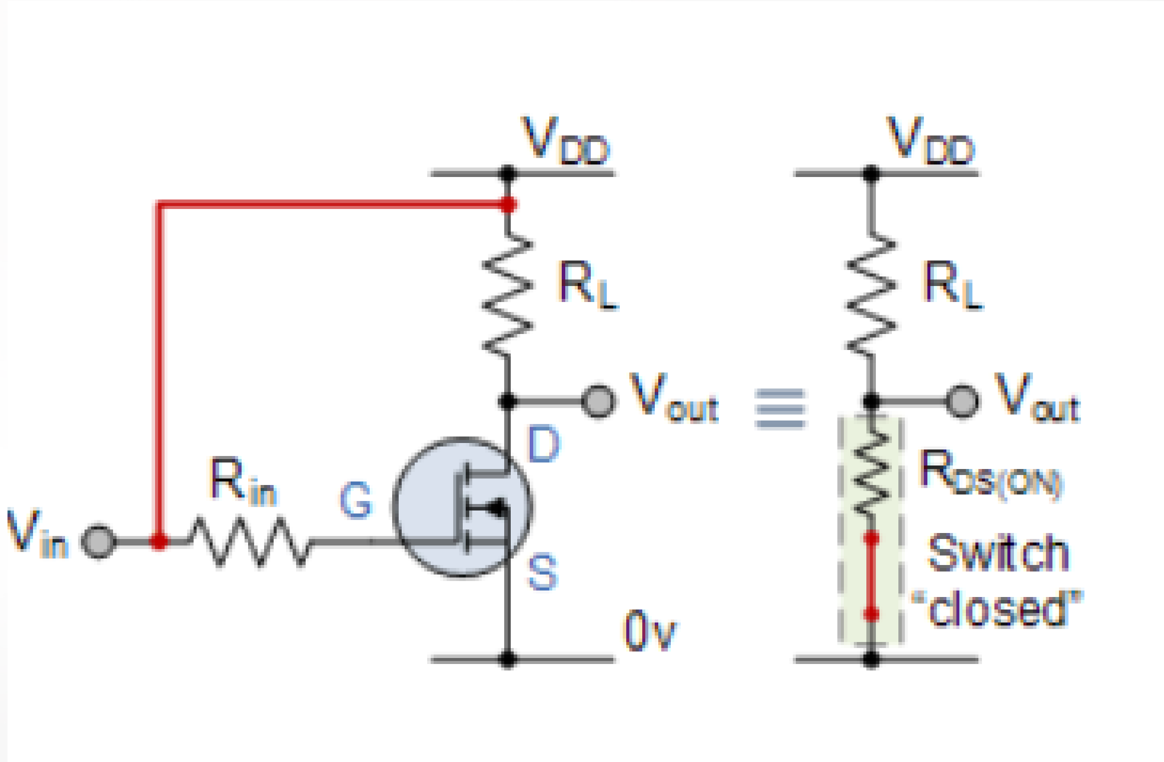




ELECTRONIC SWITCH

The core principle of the MOSFET is that the voltage applied to one terminal, the gate (G), controls the conductivity of a channel between two other terminals: the source (S) and the drain (D). This makes it a highly efficient switch, as it requires virtually no input current to control a much larger current.

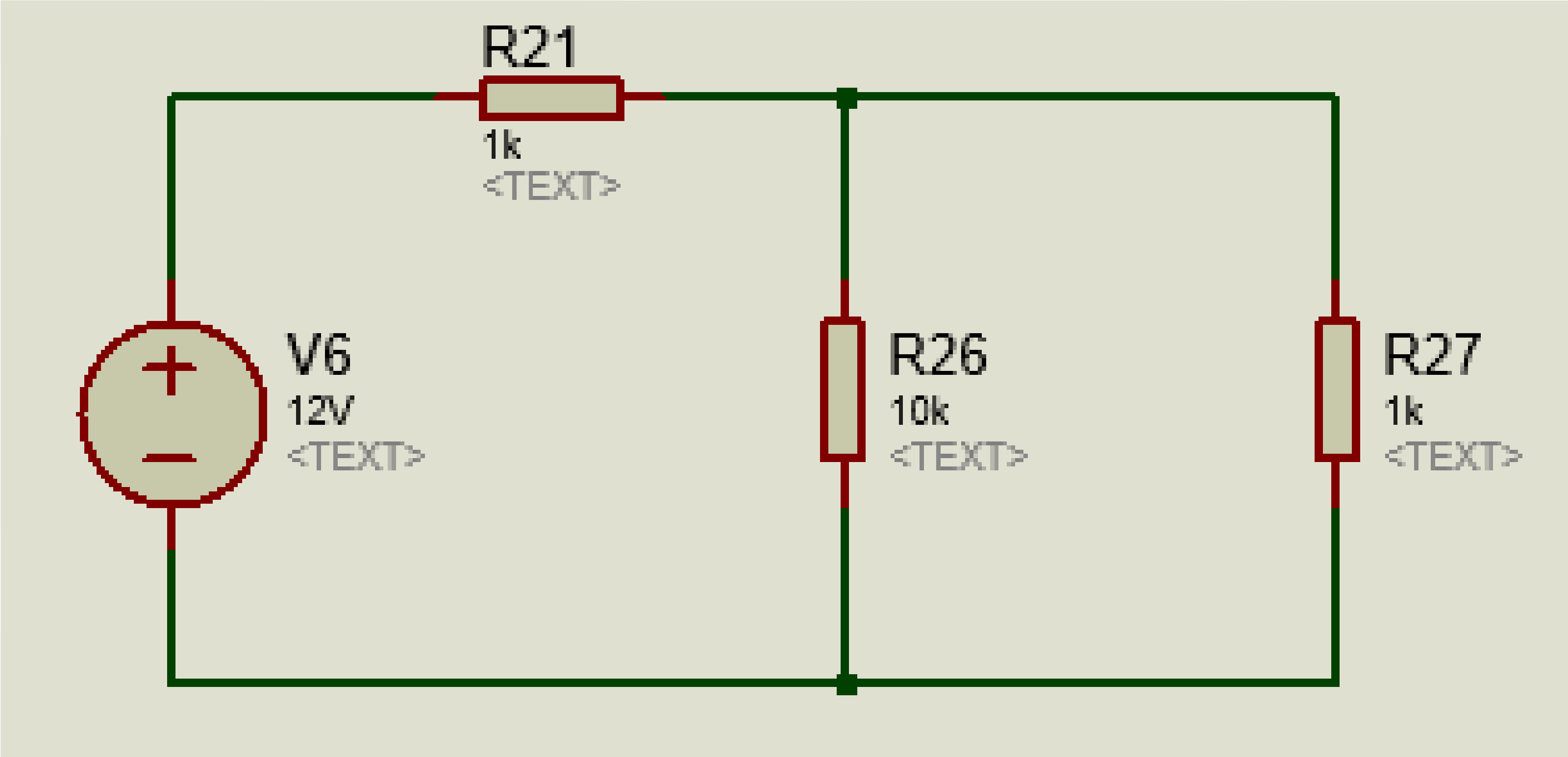




LAB1

Checking electrical characteristic.

by using multimeter, measure the electrical voltage and current which drop and cross to all resistor. record all the measured value to a table and compare to simulation.



example circuit

Multimeter

LAB2

Kirchhoff's Law

Solv by hand using Kirchhoff's law, find the voltage drop on R3 and current across R3. compare between calculation, simulation and the real circuit one.